WEEK 8 · SUBMIT PHASE · FE-11 · CHECKPOINT 2

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Date: November 2026
Platform: Google Antigravity

Production Deployment, Hygiene & Final Verification

Autonomous SEO Intelligence Platform · FlyRank Internship

PRODUCTION URL	TOTAL TEST COVERAGE	ABUSE PROTECTION	NEXT.JS 15 BUILD
`vercel.app` **(Promoted)**	**82 Automated Tests (100% Pass)**	**30 req/min · 4k char cap**	**16/16 Routes Compiled (3.7s)**

1. Production Promotion & Hygiene Overview

Assignment **FE-11** marks the final submission milestone (**Checkpoint 2**) of the FlyRank Frontend AI Engineering track. The platform is promoted to production with strict operational guardrails: streaming connection ceilings, IP-based sliding window rate limiting, prompt character caps, comprehensive documentation for recruiters and engineers, and an honest technical reflection on AI-assisted engineering workflows.

2. API Abuse Protection & Security Guardrails

Hygiene Guardrail	Configuration	HTTP Response	Operational Objective
Streaming Ceiling	`export const maxDuration = 30;`	Vercel Edge termination	Prevents zombie or hanging SSE streams from consuming edge compute budget.
IP Rate Limiter	Sliding window (30 requests / 60 seconds)	`429 Too Many Requests`	Throttles automated scrapers and brute-force token exhaustion. Emits `Retry-After` headers.
Input Character Cap	`MAX_PROMPT_CHAR_LIMIT = 4000`	`413 Payload Too Large`	Prevents prompt injection denial-of-service and extreme upstream LLM token costs.
Secret Cleanliness	`.gitignore` + `.env.example`	Zero committed secrets	Strictly guards environment keys; provides dummy mocks and placeholder configurations.

3. Cross-Browser & Viewport Compliance

Cross-Browser Verification Pass

- **Google Chrome & Chromium:** WebGL 2.0 fragment shader, SSE streaming chunks, and compositor CSS micro-interactions render at 60 FPS.

- **Mozilla Firefox:** Full adherence to standard CSS gradients, WebGL canvas buffers, and keyboard navigation rings.
- **Apple Safari (macOS & iOS 17+):** Smooth kinetic scrolling without horizontal overflow; 44px touch targets; WebGL quad shader caps DPR to 2.0 to prevent thermal throttling.
- **Screen Reader Accessibility:** Polite transcript updates (`aria-live="polite"`) eliminate screen reader stutter during real-time token delivery.

4. Production Documentation & Run Invariants

The final `README.md` has been completely updated to serve as a comprehensive onboarding guide for recruiters, tech leads, and peers:

- **What It Does:** Complete breakdown of generative UI schemas, SSE chat streaming, 3D neural nodes, GLSL fullscreen shaders, and FinOps meters.
- **60-Second Run Instructions:** Fully reproducible installation commands (`git clone` , `npm install` , `npm test` , `npm run build`).
- **Environment Variables Specification:** Table defining required vs optional variables with safe defaults.
- **How AI Tools Built This:** Technical analysis detailing where Google Antigravity excelled (GLSL mathematics, test-driven scaffold, SSR isolation) and where human engineering directed the outcome (physics dampening, contrast scrims, headless testing).

5. Checkpoint 2 Final Status Matrix

Deliverable Item	Target URL / File Reference	Status
Production Deployment	`https://flyrank-internship-front-end-ai-engineering.vercel.app`	**LIVE & VERIFIED**
Final Comprehensive README	`E:\FlyRank\Front-end AI Engineering\README.md`	**DOCUMENTED**
Abuse Protection Suite	`src/app/api/chat/route.js` & `tests/production-hygiene.test.js`	**6/6 PASS**
Full Automated Test Suite	`npm test` (82 tests across FE-01 to FE-11)	**82/82 PASS**
Next.js 15 App Router Build	`npm run build` (16/16 routes compiled)	**COMPILED (3.7s)**